Fat Cat

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Fat Cat is a saturator that keeps your dynamics. The signal is divided by its own level, saturated, then multiplied back, so it sounds the same whether you feed it quietly or loudly. Use it to add weight, warmth and bite to basses, leads, drums and buses.

Quick start

• Start from a preset, then raise Drive until the sound feels thicker, not louder.
• Use Color to pick the band that gets the most saturation (150 Hz for body, 2 to 6 kHz for bite).
• Add Transistor for warm, thick even harmonics and Grit for a lopsided fuzz-style edge; both stack on top of Drive.
• Add Fuzz for edge on the highs and Stereo Turbo for width.
• Drive character: Shipped is the plain saturation. The PurrOne stack swaps it for the three-stage colour shaper from the PurrOne synth (drive, transistor, grit), nastier above 80 % and still smooth.
• Match Output to the bypassed level while comparing.

How the window works

• Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
• Meters show the real signal in dB with a dB scale: input and output on the far right.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
• Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Saturation

Drive character	Choice: Shipped / PurrOne stack (default Shipped)	Shipped is the plain saturation (Transistor and Grit can be added on top). The PurrOne stack is the three-stage colour shaper from the PurrOne synth: opamp drive, Neve-style transistor and fuzz, nastier above 80 %, smooth (no digital fizz), with a fizz low-pass that closes as it gets nastier.
Drive	0 to 100 % default 40 %	Drive: how hard the sound is saturated. Even at 0 % a very light touch of warmth stays on by design. Dynamics are kept at any input level. With the PurrOne stack character this is the opamp-style tanh stage; above 80 % it stacks a second stage.
Transistor	0 to 100 % default 0 %	Transistor: asymmetric soft clip with even harmonics, warm and thick, not buzzy. Adds on top of Drive in Shipped, and is the middle stage of the PurrOne stack.
Grit	0 to 100 % default 0 %	Grit: lopsided fuzz-style clip, above 80 % it adds a smooth octave-up growl. Adds on top of Drive in Shipped, and is the last stage of the PurrOne stack.

Tone

Color	0 to 100 % default 0 %	Color: emphasises one band of the sound before it is saturated. Low = bass and warmth, high = bite and air.
Fuzz	0 to 100 % default 0 %	Fuzz: adds clipped high frequencies (above about 1.5 kHz) for a buzzy, raw top end.

Stereo

Stereo Turbo	0 to 100 % default 0 %	Stereo Turbo: delays the right channel by a few milliseconds for width. Check the mix in mono, too much can thin it out.

Level

Mix	0 to 100 % default 100 %	Mix: blend between the original (dry) and the saturated (wet) sound.
Output	-24 to 12 dB default 0 dB	Output: final level. Use it to match the loudness of the original when comparing.

Recommendations

- Bass: Drive 40 to 60, Color around 150 to 400 Hz, Fuzz low so the sub stays clean.
- Lead or supersaw: Drive 30 to 50, Color 2 to 4 kHz, Stereo Turbo for width; lower Mix if it gets harsh.
- Drums or bus: Drive 15 to 30 with Mix 60 to 100 for gentle glue.
- Stereo Turbo delays the right channel: check the result in mono before you commit.
- Every knob is smoothed over 20 ms, so moves are heard at once without clicks.
- A soft peak clipper (no knob) always runs just before the safety limiter: it only shaves the tallest peaks so the sound stays loud and clean.
- Latency: a few samples (oversampling), reported to your host, which compensates it.

Presets

Init, Subtle Glue, Fat Drums, Chonky Bass, Fuzzy Lead, Wide Pad, Wall Of Dirt, Colour Stack, Dirty Stack, Warm Transistor, Gritty Bass

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The picture is taken from the interactive test page of the plugin, which mirrors the plugin window; the compiled plugin may differ by a few pixels of layout.



Signal flow

Input > Saturation (Drive, Transistor, Grit) > Tone (Color, Fuzz) > Stereo Turbo > Mix and Output. Transistor and Grit only work with the PurrOne stack character.

Header bar

1	**Preset menu**	Factory presets (our own) and your saved presets.
2	**Randomize**	Set the knobs to random values (output level, mix, cuts and pan are kept).
3	**Clipper**	Soft peak clipper right before the safety limiter (always there, no knob): shaves only the tallest peaks so the limiter works less and the whole thing sits louder without harshness. Switch it off to hear the difference.
4	**Limiter**	Safety limiter on the output: keeps the level from ever passing 0 dBFS, smooth and transparent. Turn it off for the raw output.
5	**Info bubbles (i)**	Info bubbles: hover a control for one second to see what it does. Click to turn them off or on.

A. Saturation



Display A: Transfer curve of the saturation. Horizontal: normalised input level, vertical: output. The orange line shows how Drive (and the PurrOne stack stages) bend the wave; the white dot is the live level. The label top right names the character and the gain from Drive.

6	**Drive**	Drive: how hard the sound is saturated. Extended range goes to 300 %. With the PurrOne stack character this is the opamp-style tanh stage; above 80 % it stacks a second stage. (shown: 40 %)
7	**Transistor**	Transistor stage (PurrOne stack): asymmetric soft clip with even harmonics, warm and thick, not buzzy. Only active with the PurrOne stack character. (shown: 0 %)
8	**Grit (fuzz stage)**	Fuzz stage of the PurrOne stack: lopsided hard clip, above 80 % adds a smooth octave-up growl. Only active with the PurrOne stack character. (shown: 0 %)
9	**Saturation character**	Shipped is the current plain saturation. The PurrOne stack is the three-stage colour shaper from the PurrOne synth: opamp drive, Neve-style transistor and fuzz, nastier above 80 %, smooth (no digital fizz), with a fizz low-pass that closes as it gets nastier. (shown: Shipped)

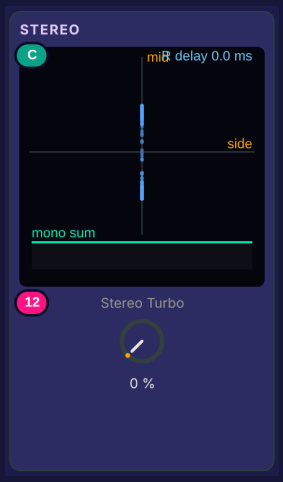
B. Tone



Display B: Tone curve: the equalisation the Color and Fuzz knobs add before and inside the saturation. Horizontal: frequency (50 Hz to 20 kHz), vertical: dB (-6 to +18). Flat at 0 dB means both knobs are off. The labels show the Color centre frequency and that Fuzz adds highs.

10	**Color**	Color: emphasises one band (150 Hz to 6 kHz) before saturation. Low = warmth, high = bite and air. (shown: 0 %)
11	**Fuzz**	Fuzz: adds clipped high frequencies (above about 1.5 kHz) for a buzzy, raw top end. Extended range goes to 300 %. (shown: 0 %)

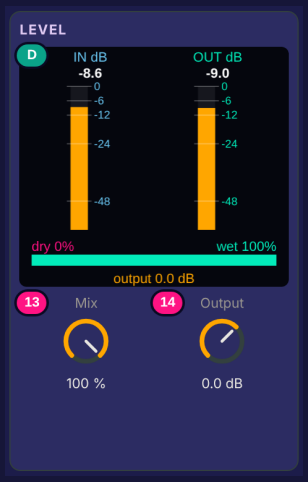
C. Stereo



Display C: Stereo Turbo view: a vertical line for the mid (centre) signal, a horizontal one for the side signal, a mono sum bar and the right channel delay in ms. With Turbo at 0 nothing is delayed.

12	**Stereo Turbo**	Stereo Turbo: delays the right channel by up to 12 ms for width. Check the mix in mono, too much can thin it out. The lab feeds the same mono source to both sides so you hear the pure effect. (shown: 0 %)

D. Level



Display D: Input and output meters in dB and the dry/wet bar (pink dry, teal wet), with the output level below.

13	**Mix**	Mix: dry/wet. (shown: 100 %)
14	**Output**	Output level. (shown: 0.0 dB)

Under the hood

• Two characters: Shipped is the plain tanh style saturation; PurrOne stack is the three stage colour shaper from the PurrOne synth (opamp tanh, asymmetric transistor with even harmonics, lopsided fuzz).
• With the stack, Drive above 80 % adds a second stage and Grit above 80 % adds a smooth octave-up growl. A fizz low-pass closes as it gets nastier so there is no digital buzz.
• Stereo Turbo delays the right channel by up to 12 ms. Always check the mix in mono: too much can thin the sound out.
• Extended ranges: Drive and Fuzz go to 300 %.
• Knob changes reach the sound within one buffer; smoothing is at most 20 ms.

Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.